

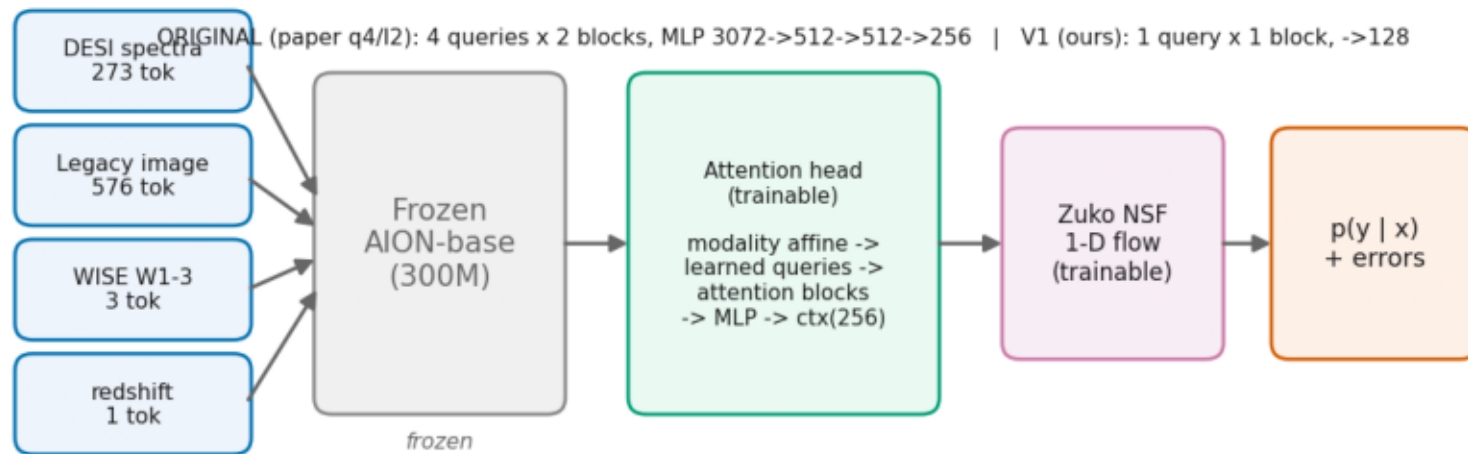
Predicting X-ray Properties from Optical/IR with AION + Normalizing Flows

Cleaned crossmatch · uncertainty-aware likelihoods · per-target flows

eROSITA eRASS1 × DESI DR1 × Legacy Survey | FASRC | extends the PAI26 paper

Architecture

Frozen AION-1 base (318M + codecs) → attention pooling → conditional NSF flow

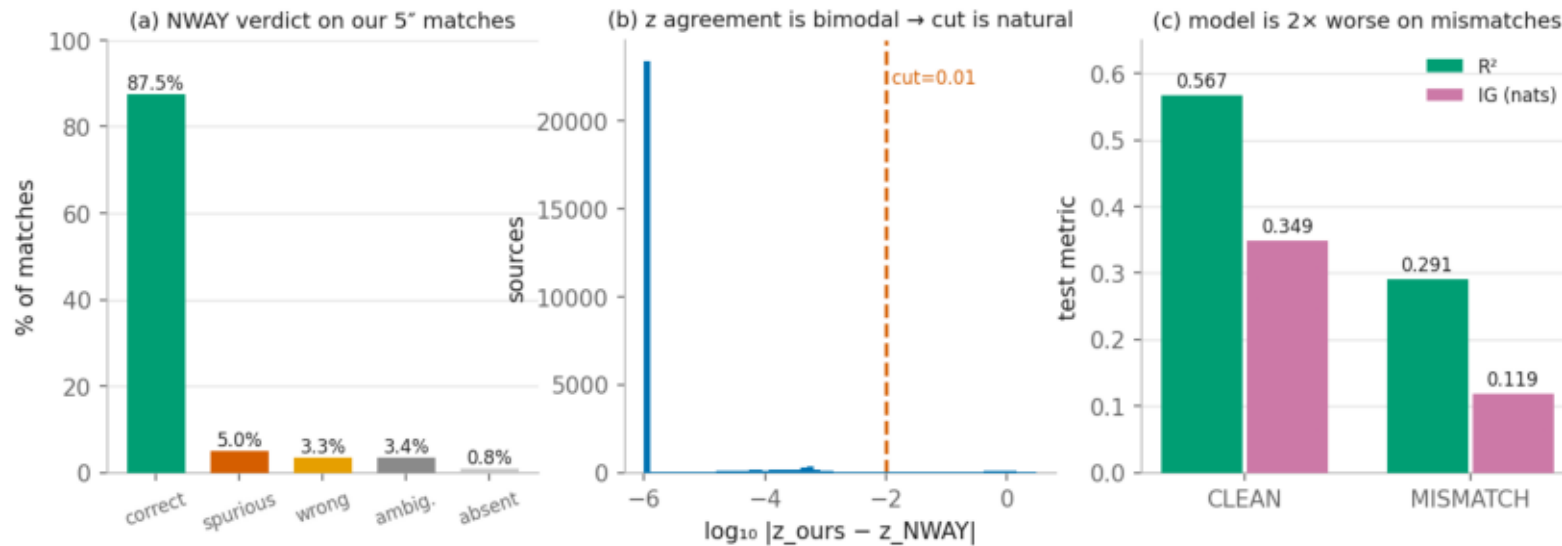


Per-batch a modality combo is sampled; AION frozen; KDE(target) prior anchors information gain.

- Per batch: one modality combo (25% singles / pairs / triples / all)
- Paper head: 4 queries, 2 layers (~16.8M)
- V1 head: 1 query, 1 layer (~7.8M)
- Flow: 1-D neural spline, KDE prior anchors info gain
- Per-target runs: flux, L_x , $\log M^*$, HR

Data & NWAY cleaning

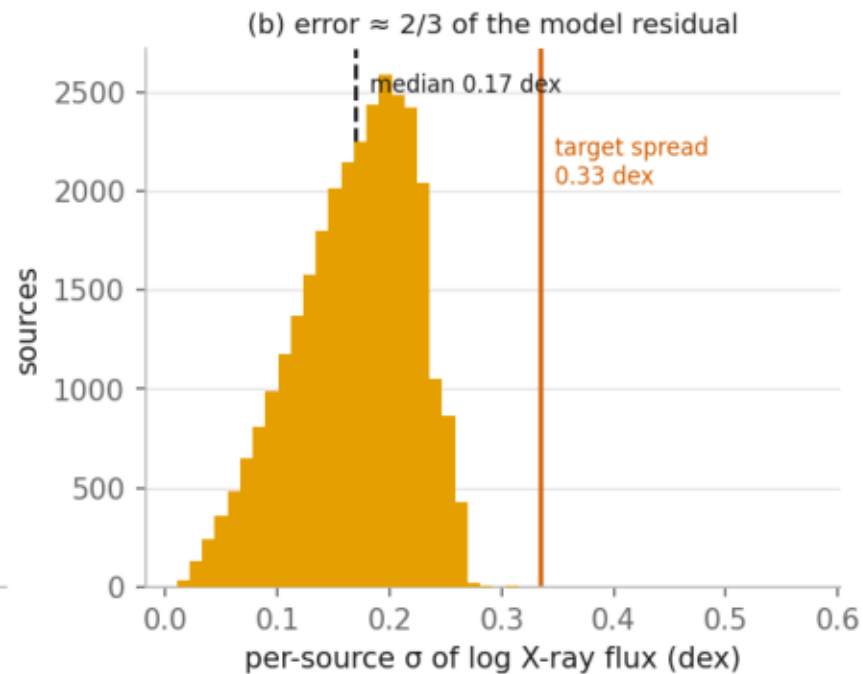
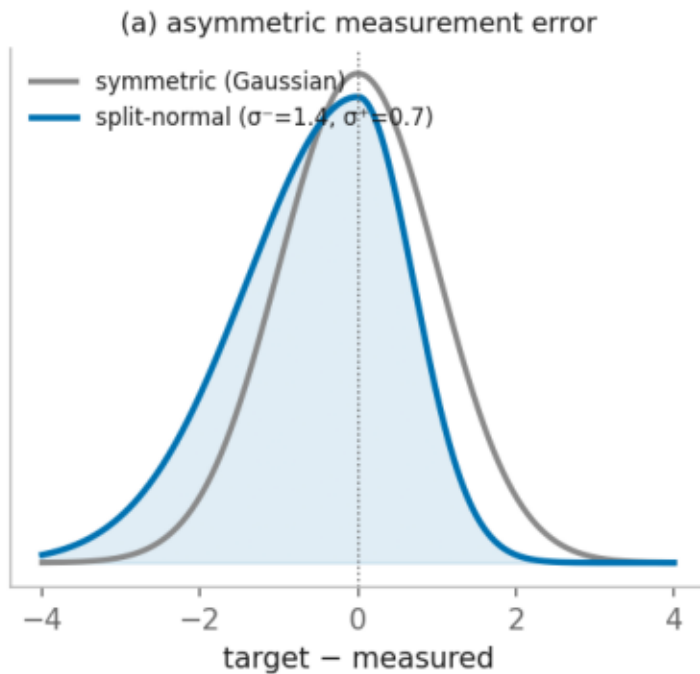
32,092 paper rows → keep NWAY-correct only → dedup → re-split (25,200)



- Naive 5" match: 12.5% not confirmed
- correct 26,632 · wrong 1,017 ·
 - ambiguous 1,033 · spurious 1,514
- Paper model 2× worse on rejects
- Impossible values ($\log L_x=48.1$)
 - are all NWAY rejects
- One staged copy + runtime view

Measurement-error-aware training

Split-normal kernel; flow learns the deconvolved density



- $\log p(y|x) = \log \int p(t|x) K(y|t) dt$
- 41-node quadrature, $\pm 5\sigma$
- flux σ : $-\log_{10}(1-LO/F), \log_{10}(1+UP/F)$
- HR: $u = \text{arctanh}(HR), \sigma_u = \sigma/(1-HR^2)$
- $\log M^*$: spectype floor 0.2/0.3 dex
- Eval scores through the same kernel

Validation gates (and what they caught)

No GPU-hour runs on unvalidated data

- `validate_staged.py`: schema · leakage · clean filter · $\sigma > 0$ · physical ranges · derived-column consistency · fits coverage · model contract — hard gate in `train.sbatch`
- Caught: interrupted unzip had silently dropped 62% of the sample (10,355/27,373 cutouts on disk; no error anywhere)
- Caught: 4 `logmstar=0` sentinel rows; out-of-range `Lx/logM*` rows all trace to NWAY-rejected matches
- Caught (by audit): convolve-trained flows were scored un-convolved — IG was structurally understated
- Break-one-thing tests: every validator check has a negative test that proves it fires
- Ops: count-based idempotency · sbatch-only compute · one canonical data copy · measured timings before submits

Results — log flux (0.2–2.3 keV), cleaned + convolve

Anchors: paper published 0.549 (noisy test) · paper model on clean rows 0.567

V1 head (7.8M, 15 epochs)

inputs	R ²	exp(IG)	RMSE (dex)
z	0.199	0.98	0.295
wise	0.356	1.09	0.264
image	0.455	1.15	0.243
spectra	0.512	1.14	0.23
spectra+z+imag	0.572	1.23	0.215
spectra+z+wise+im	0.569	1.23	0.216

- Modality ordering reproduces the paper: z < WISE < image < spectra < combos
- QSO R² 0.60 (n=2,195) vs GALAXY 0.40 (n=325); signal concentrated at z < 0.7

Next steps

- Finish V1 vs paper-head A/B on identical clean data (in flight) → pick the sweep head
- Paper-table reproduction (paper head, error_mode=none, native split, 50 epochs)
- Per-target runs: $\log L_x \cdot \log M^* \cdot \text{HR}(\text{arctanh}, \text{IG-primary}, \text{S/N gate})$
- AION token cache (~185 GB per-combo): 5–15× epoch speedup for the sweep
- siag_gpu (32×A100) once membership lands; V2 self-trained encoder; upper-limit censoring